



# Digi Twin

A live, explainable Cardiac Digital Twin

Digital Twin Challenge 2026 | Happiest Health

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# The problem

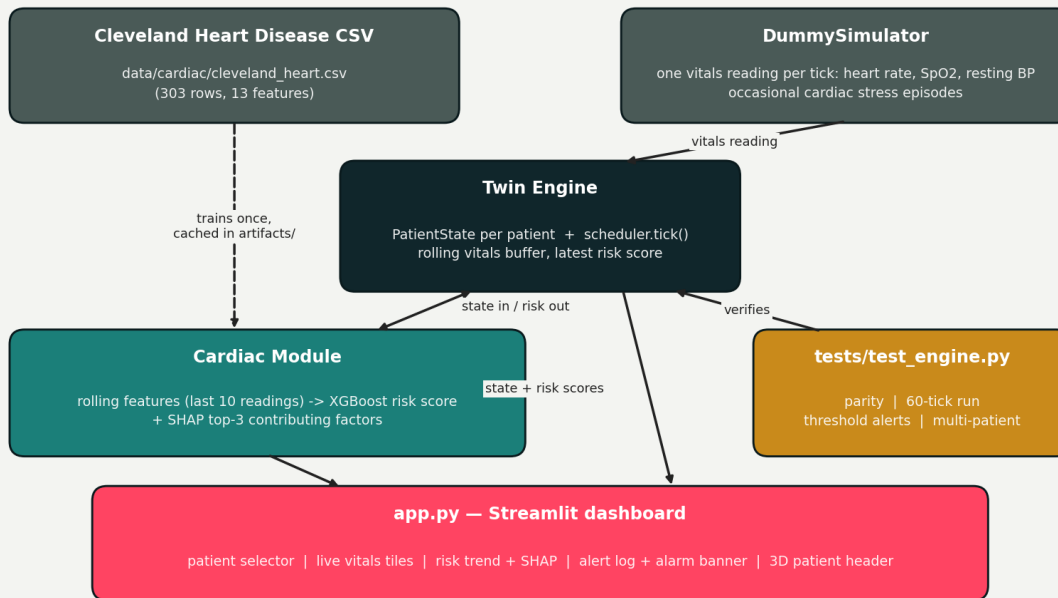
- Most cardiac risk tools score a patient once, from a single snapshot.
- Risk that develops between visits is missed.
- A bare number rarely explains why risk is high, so it is hard to trust or act on.
- Wearables produce a steady stream of vitals, but they are rarely joined to the patient's health record.

# Our solution and healthcare use case

- One evolving digital twin per patient: EHR baseline plus a rolling window of live vitals.
- Cardiac risk is re-scored on every new reading, so risk becomes a continuously updated line.
- Each score ships with SHAP factors showing what drove it.
- Alerts when risk crosses the threshold: an alert log and a flashing alarm banner.
- Multi-patient dashboard, each patient with independent state.

# Architecture

## Digi Twin — Cardiac Digital Twin Architecture



# AI/ML model

## XGBoost

200 trees, depth 5, learning rate 0.05, trained on the UCI Cleveland Heart Disease data (13 features)

## AUC-ROC ~0.92

Held-out score on an 80/20 stratified split, features standardised

## SHAP

TreeExplainer on every tick; dashboard shows the top three contributing factors

## Rolling features

Resting BP averaged and max heart rate taken over the last 10 readings; other fields from the EHR baseline

# Technical stack and features

## Stack

- Python 3.12
- Streamlit and Plotly
- XGBoost, scikit-learn
- SHAP, pandas, NumPy
- Web Bluetooth
- flake8, tests, GitHub Actions CI

## Features

- Live risk re-scored every tick
- Explainable SHAP factors
- Realistic simulator with stress episodes
- Edge-triggered alert log and alarm banner
- Live BLE heart-rate mode in the browser
- 3D patient figure and vital tiles

# Validation, limitations and roadmap

## Validation

- Model parity check
- 60-tick multi-day consistency
- Threshold-breach alerts
- Multi-patient personalization
- CI runs lint and tests on every push

## Limitations and roadmap

- Small training set (about 300 rows); vitals are simulated
- Not for clinical use
- Next: real EHR and wearable feeds, larger datasets, more organ systems

# Team

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B.Tech Mechanical, 2027

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[github.com/harsh23533-del/digital-twin-](https://github.com/harsh23533-del/digital-twin-)